Reinforcement Learning vs. Traditional Portfolio Strategies During Market Downturns

A cost-aware comparison across the Global Financial Crisis, COVID-19, and 2022

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5 strategies • 3 historical events • 9 U.S. sector ETFs • $100,000 initial capital

RESEARCH QUESTION

- **How do PPO and DDPG compare** with traditional portfolio strategies during market downturns and recovery?
- Measure net return, risk, transaction costs, and recovery using the same assets and cost-aware backtest.
- Compare Equal Weight, Minimum Variance, Mean Variance, PPO, and DDPG; no strategy is assumed to win.

DATA & EVENT DESIGN

- Nine U.S. sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY); adjusted-close returns from Yahoo Finance. Cboe VIX is a market-state feature, not an asset.
- Daily coverage: 2003–2024; 252-session warm-up for rolling strategies.

Event	Train	Validate	Downturn test	Recovery observed
GFC	04/2004–10/2006	10/2006–10/2007	10/2007–03/2009	03/2009–03/2011
COVID	02/2016–02/2019	02/2019–02/2020	02/2020–03/2020	03/2020–03/2022
2022	01/2018–01/2021	2021	01/2022–10/2022	10/2022–10/2024

Calendar ranges shown; calculations use inclusive, aligned trading-day endpoints (last data: 2024-10-11).

SHARED DECISION & EVALUATION PIPELINE

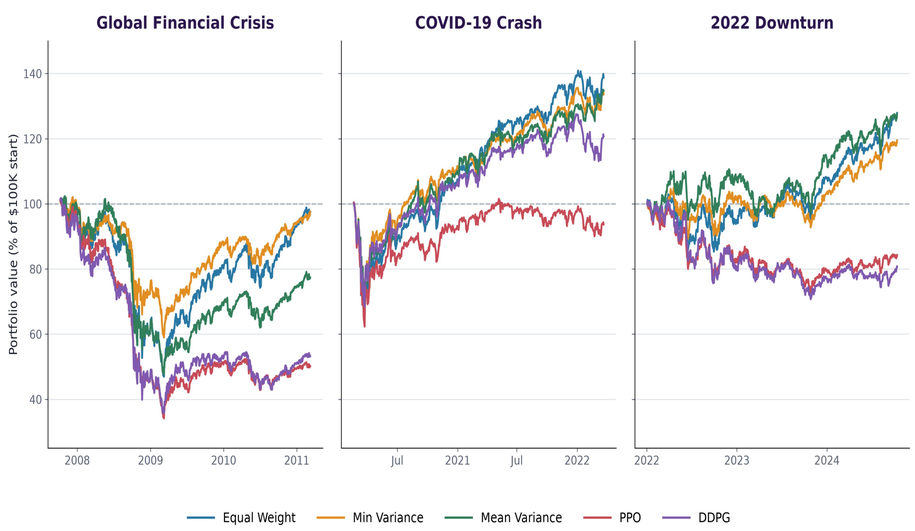
INPUTS	20 days of 9 ETF returns + lagged VIX, VIX change, and current weights
191-D STATE	Lagged information only; no same-day closing-return lookahead
POLICY	Equal Weight • MinVar • MVO • PPO • DDPG → target allocation
TRADE & SCORE	Long-only capped weights → fees → net return, risk, and recovery

BACKTEST & TRAINING CHOICES

- **Start at $100K;** fully invested, long-only, 30% maximum per ETF. Equal Weight rebalances monthly; optimizers and RL act daily.
- Decisions use information available after the prior close; the allocation earns the current day's return.
- **Turnover =** sum of absolute weight changes. Fee = 0.10% × pre-trade value × turnover; charged once before return.
- PPO/DDPG: 100,000 requested steps per model; seed 42; best checkpoint selected on validation. Observation normalization fitted on training data only.
- 252 trading days/year; zero daily risk-free rate for Sharpe (explicit simplifying assumption).

PORTFOLIO PATHS THROUGH DOWNTURN & RECOVERY

Daily portfolio paths: downturn plus observed recovery



Daily portfolio value as a percentage of the $100K starting capital; transaction costs included. Dashed line = starting capital. Curves extend through each observed recovery window.

DOWNTURN PERFORMANCE

Net total return / maximum drawdown during the downturn (costs included)

Strategy	GFC Return / MDD	COVID Return / MDD	2022 Return / MDD
Equal Weight	-53.1% / -53.5%	-36.5% / -36.7%	-14.1% / -17.0%
Min Variance	**-41.0% / -42.2%**	**-29.7% / -29.7%**	-9.7% / -14.8%
Mean Variance	-52.4% / -53.4%	-31.4% / -31.4%	**-2.6% / -13.5%**
PPO	-65.8% / -66.1%	-37.7% / -37.9%	-22.0% / -23.3%
DDPG	-64.2% / -64.9%	-31.5% / -31.5%	-22.8% / -25.2%

Best-return cells highlighted. All 15 strategy/event downturn returns were negative; lower loss is better.

TAKEAWAY

No universal winner—and no RL strategy led a tested downturn.

- Minimum Variance had the strongest downturn return and Sharpe in the 2007–09 and 2020 tests.
- Mean Variance led in 2022, with a −2.6% net downturn return versus −22.0% PPO and −22.8% DDPG.
- These are historical, single-seed results—not evidence of general or statistically significant superiority.

WHAT STANDS OUT?

- **Traditional portfolios led** the best downturn return in all three events: MinVar in GFC/COVID; MVO in 2022.
- **RL strategies** traded more dynamically and incurred much higher fees in GFC and 2022.
- Downturn fees — GFC: PPO $17.8K; DDPG $20.1K. 2022 fees: PPO $12.1K; DDPG $13.2K.
- Frequent reallocation is not automatically protective when turnover costs compound.

RECOVERY TO $100K

Strategies first crossing initial capital after the downturn trough:

GFC	0 / 5	None recovered within 24 months
COVID	5 / 5	All crossed; PPO later ended below $100K
2022	3 / 5	PPO and DDPG did not recover

Recovery = first crossing of $100K after trough; a temporary crossing counts.

LIMITATIONS

- One random seed (42) and one validation episode per checkpoint; no confidence intervals or multi-seed robustness.
- Only nine U.S. sector ETFs and three selected historical windows; results may not generalize.
- Constant 10 bp trading cost and zero daily risk-free rate are simplifying assumptions.
- Recovery is observed only through the available window (data end: 2024-10-11).

SOURCES & REPRODUCIBILITY

- Markowitz, H. (1952). Portfolio Selection. Journal of Finance, 7(1), 77–91.
- Schulman et al. (2017). Proximal Policy Optimization Algorithms. arXiv:1707.06347.
- Lillicrap et al. (2015). Continuous control with deep reinforcement learning. arXiv:1509.02971.

- Data: Yahoo Finance (yfinance) adjusted closes; Cboe VIX history.
- Run settings: configs/experiment.yaml • seed 42 • Stable-Baselines3 2.9.2.
- Daily results: results/tables/summary_metrics.csv; full pipeline: src/experiments/run_experiment.py.